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VoltGuard

Engenharia e Gestão de Serviços

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The Problem

Global studies and industrial audits report that industrial facilities waste up to 30% of energy due to lack of visibility

- No **real-time** visibility into energy
- Anomalies detected **too late**, after damage is done
- No **centralized** control across devices and sites

Lack Of Real-Time Energy Visibility

Here's where things get interesting.

Without **real-time energy monitoring**, making informed decisions about energy consumption is like driving blindfolded. It's dangerous and ineffective.

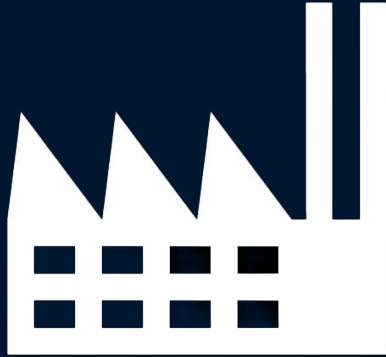
Manufacturers with traditional energy audits typically achieve only 5% energy savings, while those implementing real-time monitoring systems can slash energy consumption by up to 20% [4].

Meet VoltGuard

- Real-time monitoring across all connected devices
- Automatic anomaly detection, catch the problems instantly
- Alerts delivered before damage is done



Who is it for?



Energy Managers
Full visibility into
consumption,
costs, and trends



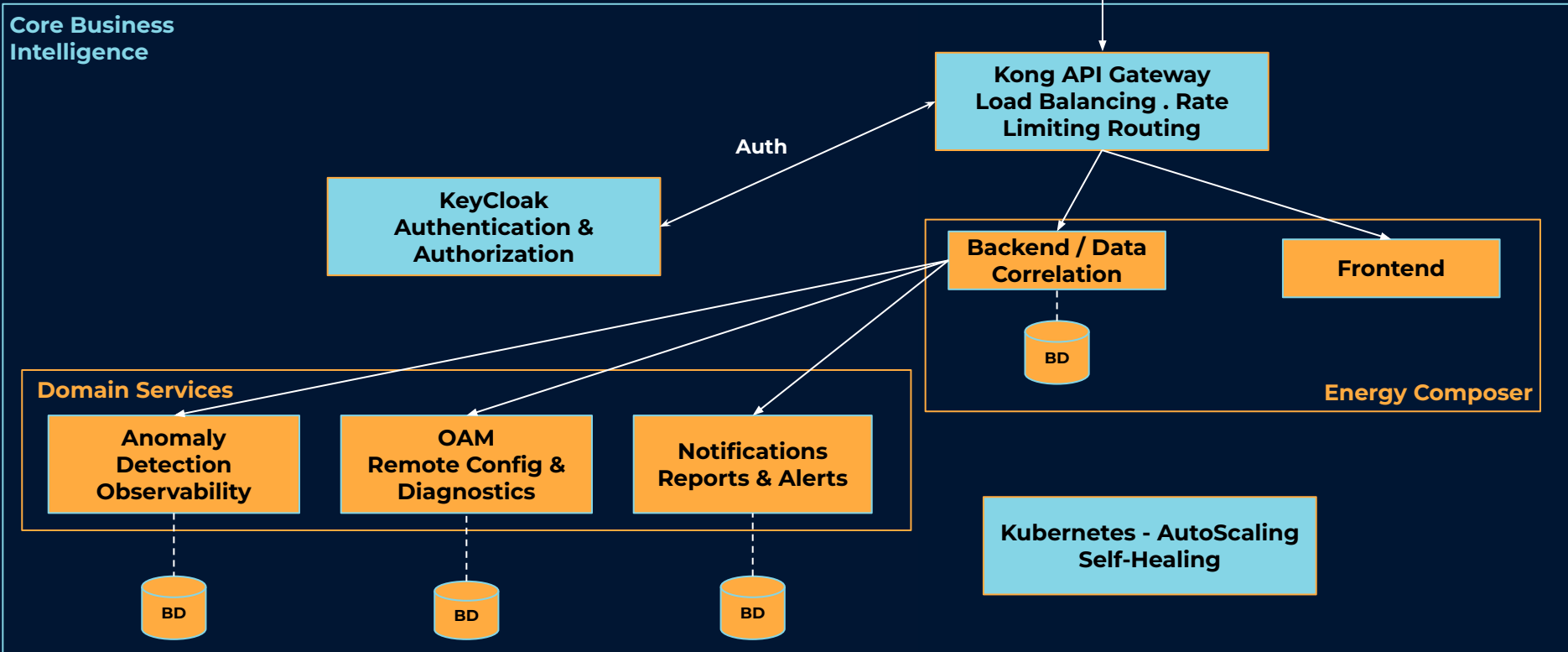
Operations Teams
Real-time alerts,
zero downtime
tolerance



Smart Facilities
Factories, Data
Centers, Buildings
with IoT
infrastructure

System Architecture

Built for Production



Walkthrough

<https://grupo4-egs-deti.ua.pt/>

Performance under load

```
diogozecca@diogolap-asusg15:~$ ab -n 100000 -c 200 http://grupo4-egs-deti.ua.pt/
This is ApacheBench, Version 2.3 <$Revision: 1903618>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/
```

```
Benchmarking grupo4-egs-deti.ua.pt (be patient)
Completed 10000 requests
Completed 20000 requests
Completed 30000 requests
Completed 40000 requests
Completed 50000 requests
Completed 60000 requests
Completed 70000 requests
Completed 80000 requests
Completed 90000 requests
apr_socket_recv: Connection timed out (110)
Total of 99978 requests completed
```

```
Every 2.0s: kubectl get pods -n tenant-grupo4-egs-deti-ua-pt -l app=grupo4-kong diogolap-asusg15: Tue Jun 2 15:57:22 2026
```

NAME	READY	STATUS	RESTARTS	AGE
grupo4-kong-69f799668f-4zn2c	0/1	Running	0	2s
grupo4-kong-69f799668f-7g58g	1/1	Running	0	25m
grupo4-kong-69f799668f-mldgc	0/1	ContainerCreating	0	2s
grupo4-kong-69f799668f-qhx56	0/1	ContainerCreating	0	2s
grupo4-kong-69f799668f-vxvbh	1/1	Running	0	11m

```
Every 2.0s: kubectl get pods -n tenant-grupo4-egs-deti-ua-pt -l app=grupo4-c... diogolap-asusg15: Tue Jun 2 15:57:20 2026
```

NAME	READY	STATUS	RESTARTS	AGE
grupo4-compositor-frontend-6fd86ffd47-4xmxd	1/1	Running	0	27s
grupo4-compositor-frontend-6fd86ffd47-9vf8s	1/1	Running	0	25h
grupo4-compositor-frontend-6fd86ffd47-gsp52	1/1	Running	0	27s
grupo4-compositor-frontend-6fd86ffd47-j4tk9	1/1	Running	0	25h
grupo4-compositor-frontend-6fd86ffd47-x82kv	0/1	Running	0	12s

```
Every 2.0s: kubectl get hpa -n tenant-grupo4-egs-deti-ua-pt
```

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS	REPLICAS	AGE
compositor-frontend-hpa	Deployment/grupo4-compositor-frontend	cpu: 359%/50%	2	5	4	8d
kong-hpa	Deployment/grupo4-kong	cpu: 393%/75%	2	5	2	8d

```
diogolap-asusg15: Tue Jun 2 15:57:21 2026
```

```
diogozecca@diogolap-asusg15:~$ kubectl get events -n tenant-grupo4-egs-deti-ua-pt --field-selector involvedObject.kind=HorizontalPodAutoscaler --watch
```

LAST SEEN	TYPE	REASON	OBJECT	MESSAGE
13m	Normal	SuccessfulRescale	horizontalpodautoscaler/compositor-frontend-hpa	New size: 5; reason: cpu resource utilization (percentage of request) above target
11m	Normal	SuccessfulRescale	horizontalpodautoscaler/compositor-frontend-hpa	New size: 2; reason: All metrics below target
13m	Normal	SuccessfulRescale	horizontalpodautoscaler/kong-hpa	New size: 5; reason: cpu resource utilization (percentage of request) above target
10m	Normal	SuccessfulRescale	horizontalpodautoscaler/kong-hpa	New size: 4; reason: All metrics below target
9m50s	Normal	SuccessfulRescale	horizontalpodautoscaler/kong-hpa	New size: 3; reason: All metrics below target
8m35s	Normal	SuccessfulRescale	horizontalpodautoscaler/kong-hpa	New size: 2; reason: All metrics below target
0s	Normal	SuccessfulRescale	horizontalpodautoscaler/kong-hpa	New size: 3; reason: cpu resource utilization (percentage of request) above target
0s	Normal	SuccessfulRescale	horizontalpodautoscaler/compositor-frontend-hpa	New size: 5; reason: cpu resource utilization (percentage of request) above target
0s	Normal	SuccessfulRescale	horizontalpodautoscaler/kong-hpa	New size: 5; reason: cpu resource utilization (percentage of request) above target
0s	Normal	SuccessfulRescale	horizontalpodautoscaler/compositor-frontend-hpa	New size: 2; reason: All metrics below target
0s	Normal	SuccessfulRescale	horizontalpodautoscaler/kong-hpa	New size: 4; reason: All metrics below target

Resilienc

e

diogozeca@diogolap-asusg15: ~

Every 2.0s: kubectl get pods -n t... diogolap-asusg15: Tue Jun 2 17:16:42 2026

NAME	READY	STATUS	RESTARTS	AGE
grupo4-compositor-frontend-6fd86ffd47-8ztm6	1/1	Running	0	5m22s
grupo4-compositor-frontend-6fd86ffd47-d896t	1/1	Running	0	4m13s

diogozeca@diogolap-asusg15: ~

```
diogozeca@diogolap-asusg15:~$ kubectl delete pod grupo4-compositor-frontend-6fd86ffd47-8ztm6 -n tenant-grupo4-egs-deti-ua-pt
```

diogozeca@diogolap-asusg15: ~

Every 2.0s: kubectl get pods -n t... diogolap-asusg15: Tue Jun 2 17:16:49 2026

NAME	READY	STATUS	RESTARTS	AGE
grupo4-compositor-frontend-6fd86ffd47-8ztm6	1/1	Terminating	0	5m29s
grupo4-compositor-frontend-6fd86ffd47-d896t	1/1	Running	0	4m20s

diogozeca@diogolap-asusg15: ~

```
diogozeca@diogolap-asusg15:~$ kubectl delete pod grupo4-compositor-frontend-6fd86ffd47-8ztm6 -n tenant-grupo4-egs-deti-ua-pt
pod "grupo4-compositor-frontend-6fd86ffd47-8ztm6" deleted from tenant-grupo4-egs-deti-ua-pt namespace
```

diogozeca@diogolap-asusg15: ~

Every 2.0s: kubectl get pods -n t... diogolap-asusg15: Tue Jun 2 17:17:02 2026

NAME	READY	STATUS	RESTARTS	AGE
grupo4-compositor-frontend-6fd86ffd47-d896t	1/1	Running	0	4m33s
grupo4-compositor-frontend-6fd86ffd47-knpfw	1/1	Running	0	13s

diogozeca@diogolap-asusg15: ~

```
diogozeca@diogolap-asusg15:~$ kubectl delete pod grupo4-compositor-frontend-6fd86ffd47-8ztm6 -n tenant-grupo4-egs-deti-ua-pt
pod "grupo4-compositor-frontend-6fd86ffd47-8ztm6" deleted from tenant-grupo4-egs-deti-ua-pt namespace
diogozeca@diogolap-asusg15:~$
```

FIM

